

XLOOKUP Function in Excel

1. What is XLOOKUP?

The XLOOKUP function searches for a value in one column (or row) and returns the matching value from another column. It solves the problem of finding related information, such as the department of an employee or the price of a product, using an ID. A Data Analyst uses it to pull data from one table into another without searching manually.

2. Syntax

`=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])`

Argument	Required?	What it accepts
lookup_value	Required	The value to find (for example "E103" or a cell like A6)
lookup_array	Required	The single column or row to search in (for example A2:A5)
return_array	Required	The column or row that holds the result (for example C2:C5). It must be the same size as lookup_array.
if_not_found	Optional	What to show if no match exists, such as "Not Found". Without it, Excel shows #N/A.
match_mode	Optional	0 exact match (default), -1 exact or next smaller, 1 exact or next larger, 2 wildcard
search_mode	Optional	1 search first to last (default), -1 search last to first

Beginners usually need only the first three arguments, and the fourth when missing values are possible.

3. Simple Example

Find the department of Employee E103 (typed in A6):

	A	B	C
1	Employee ID	Employee Name	Department
2	E101	Aisha	Marketing
3	E102	Rahim	Finance
4	E103	Nabila	Analytics
5	E104	Tanvir	Operations
6	E103	Result	Analytics

Formula in C6: `=XLOOKUP(A6, A2:A5, C2:C5)`

Excel finds E103 in A2:A5 (it is in row 4), then returns the value from the same row in C2:C5. **Result: Analytics.** You can also type the ID directly: `=XLOOKUP("E103", A2:A5, C2:C5)`.

4. Another Example

Find the price of Product P203, with a message if the ID is missing:

	A	B	C
1	Product ID	Product	Price
2	P201	Notebook	120
3	P202	Pen	25
4	P203	Stapler	350
5	P204	Marker	90
6	P203	Result	350

Formula in C6: `=XLOOKUP(A6, A2:A5, C2:C5, "Not Found")`

Excel finds P203 in A2:A5 and returns the price from the same row. **Result: 350.** If A6 contained P999, which is not in the list, Excel would show "Not Found" instead of #N/A.

5. When to Use It

- Find employee details, such as department, using an employee ID
- Retrieve product prices or names using a product ID
- Pull customer information, such as city or segment, using a customer ID
- Match sales or orders to details stored in another table
- Check which IDs from one list are missing from another list

6. Important Things to Remember

- XLOOKUP finds an exact match by default. You do not need an extra setting for it.
- lookup_array and return_array must be the same size, and each should be one column (or one row).
- The result column can be to the left or right of the lookup column. There are no column numbers to count.
- If there are duplicate values in the lookup column, XLOOKUP returns the first match.
- If nothing matches and if_not_found is not used, the result is #N/A. Matching is not case-sensitive.
- XLOOKUP is available in Excel 2021, Excel 2024, and Microsoft 365, but not in Excel 2019 or earlier.

7. Common Mistakes

Mistake: Using arrays of different sizes, such as A2:A5 with C2:C6. **Fix:** Select the same rows for both.

Mistake: Selecting the whole table as the lookup or return array. **Fix:** Select only one column for each array.

Mistake: Typing a text lookup value without quotation marks, such as E103. **Fix:** Write "E103", or put the value in a cell and refer to that cell.

Mistake: Getting #N/A for an ID that looks correct, because of extra spaces or numbers stored as text. **Fix:** Clean the data so both lists use the same format.

8. Data Analyst Tip

When you bring data from one table into another, use the if_not_found argument with a clear message such as "Not Found", then filter that column. This quickly shows IDs that are missing from the source table. Put the lookup value in a cell and lock the arrays with \$, for example \$A\$2:\$A\$5, so you can copy the formula down.